



Ministry of Environment
and Climate Change



National Agency Meteorology
and Environmental Monitoring



Information and Research Institute of
Meteorology, Hydrology and Environment (IRIMHE)

Overview of the National remote sensing center

Lkhagvadorj Nanzad
Satellite Application and Remote
Sensing Research Division, IRIMHE



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<http://icc.mn/>



REMOTE SENSING PRODUCT

NATIONAL REMOTE SENSING CENTER

MODIS

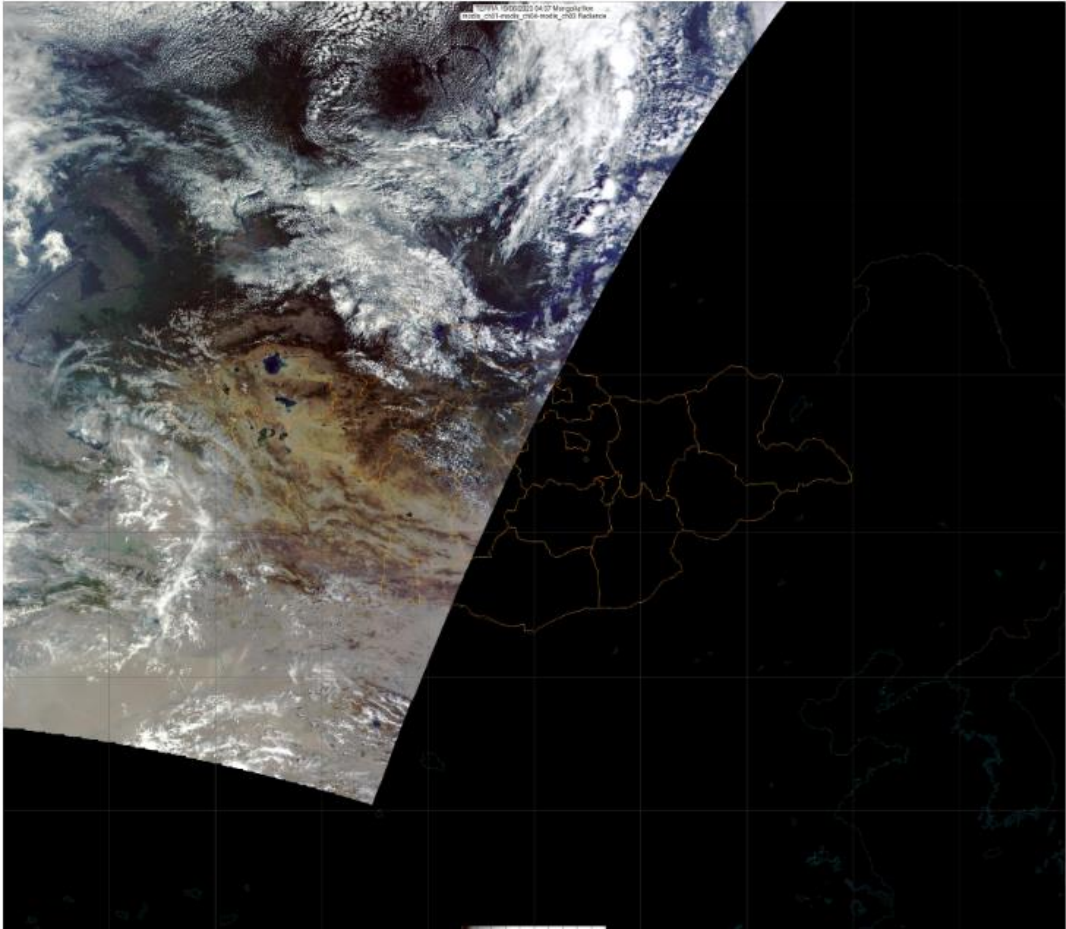
NOAA - METOP

FY

NPP - JPSS

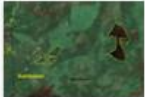
HIMAWARI

TODAY: MONGOLIA FROM MODIS SATELLITE

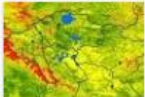


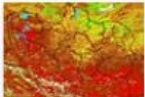
REMOTE SENSING PRODUCT

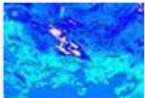
LAND

Fire

Vegetation

NDVI Anomaly

Drought

Land Surface Temperature

Snow cover

ATMOSPHERE

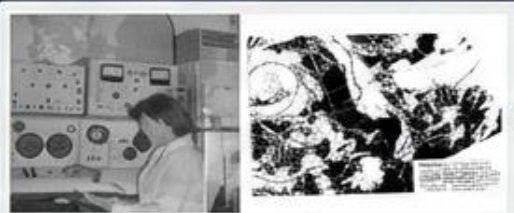
Dust

DECADES OF REMOTE SENSING HERITAGE

LAT: 47.9166° N
LONG: 186.8833° E

1970

ESSA-8 & METEOR-2. First receiving station via APT analog. Manual photograph processing for cloud movement.



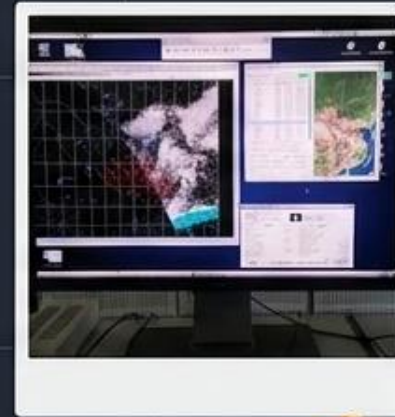
1996

NOAA Fleet (15, 17, 18, 19). Achieved 1km resolution for tracking snow cover and biomass.



1996

NOAA Fleet (15, 17, 18, 19). Achieved 1km resolution for tracking snow cover and biomass.



2007

Terra/Aqua MODIS & FY2D. Launch of 250m/500m resolution tracking for wildfires, droughts, and land surface temperature.

2012

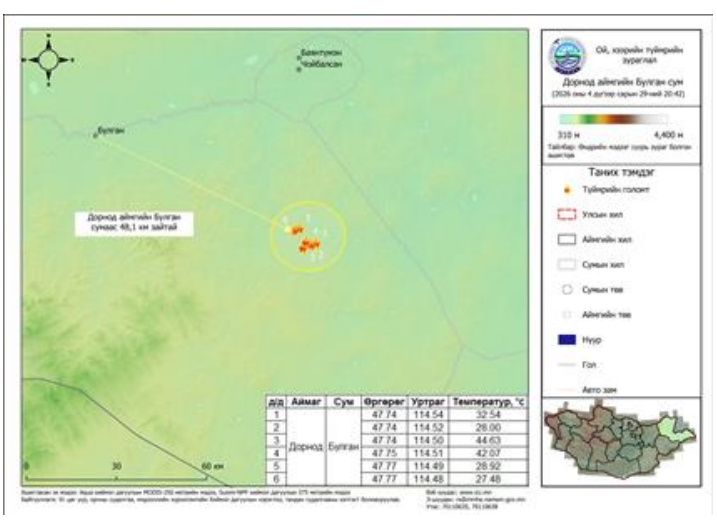
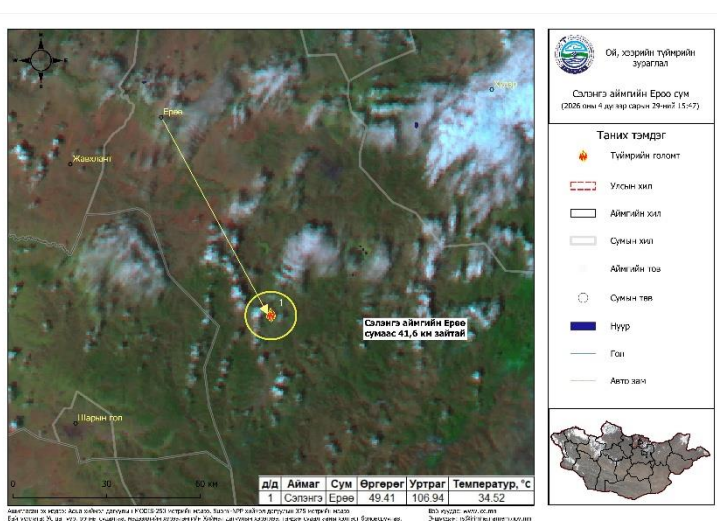
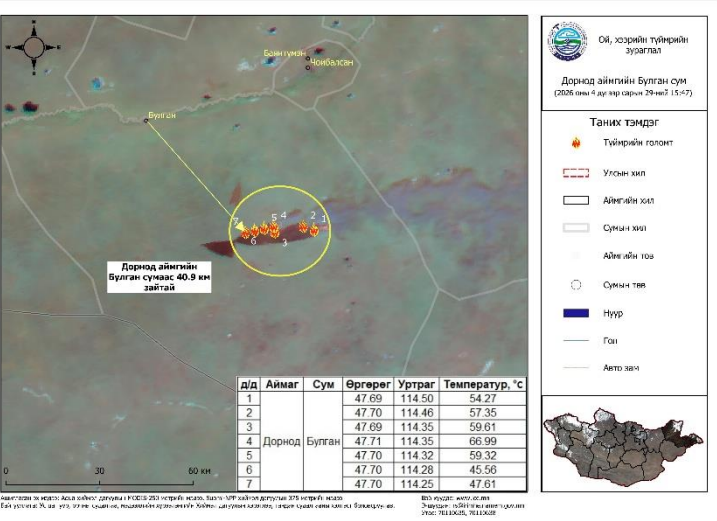
Suomi-NPP VIIRS. Integration of 750m resolution data utilizing advanced VIIRS, ATMS, and OMPS instruments.



We receive and analyze real-time data and images from Polar Orbiting Satellites such as the AQUA, TERRA, VIIRS, NOAA series, and the Geostationary Meteorological satellite - FY2C.

The satellite data are used for service as determining and imaging the actual status of snow and vegetation cover, forest and steppe wildfires.

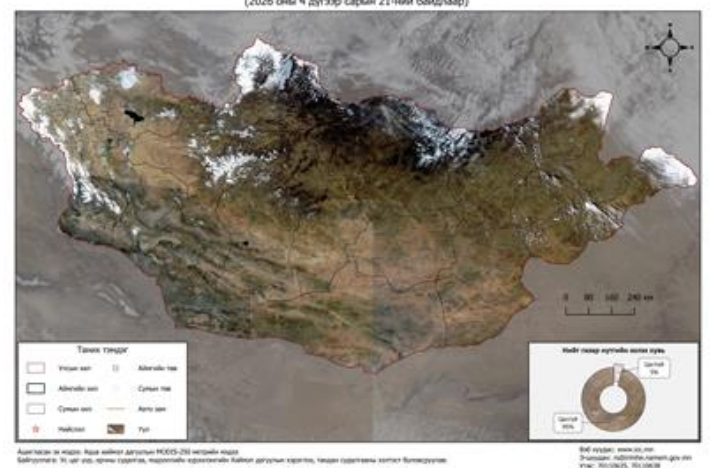
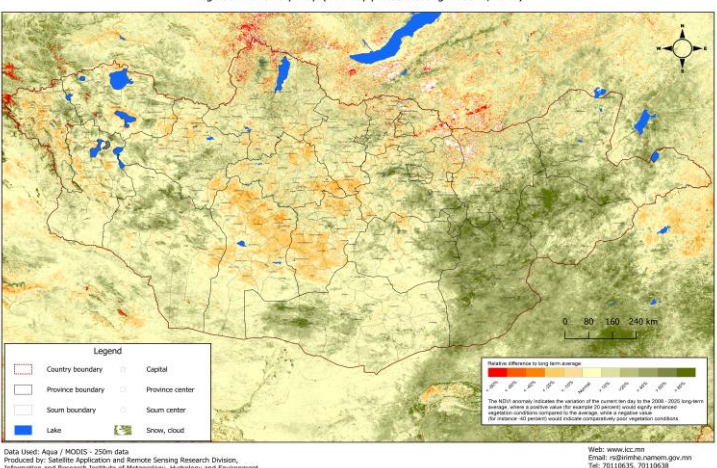
Wildfire monitoring



NDVI

NDVI anomaly

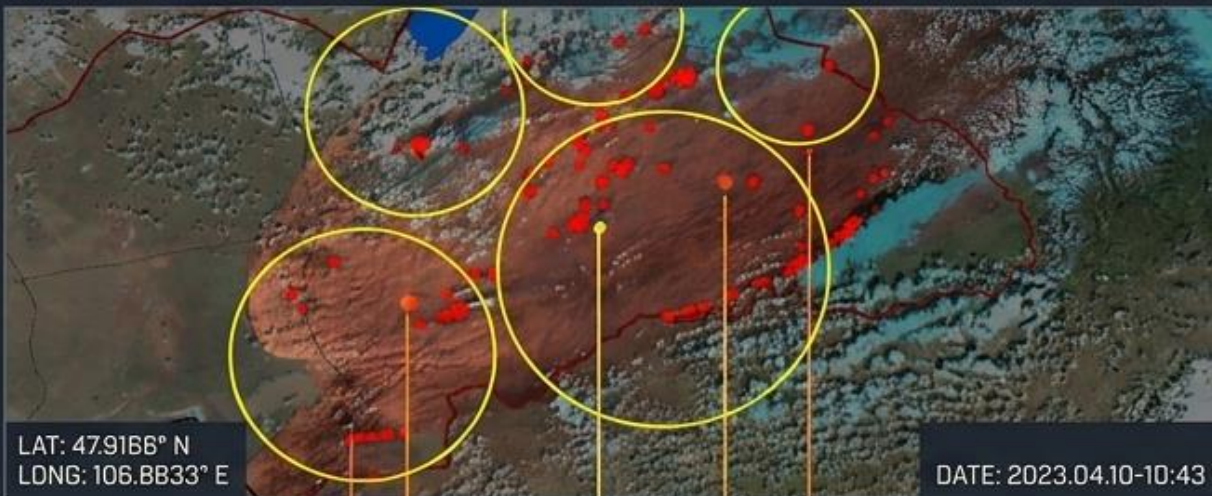
Snow cover mapping



ANATOMY OF A DISASTER RESPONSE: WILDFIRES

SATELLITE FIRE MAP - HOTSPOTS & BURN ZONES

DATE: 2023.04.10-10:43



DORNOD PROVINCE IGNITION

IMPACT: 23,114 sq.km burned area detected [April 2023].

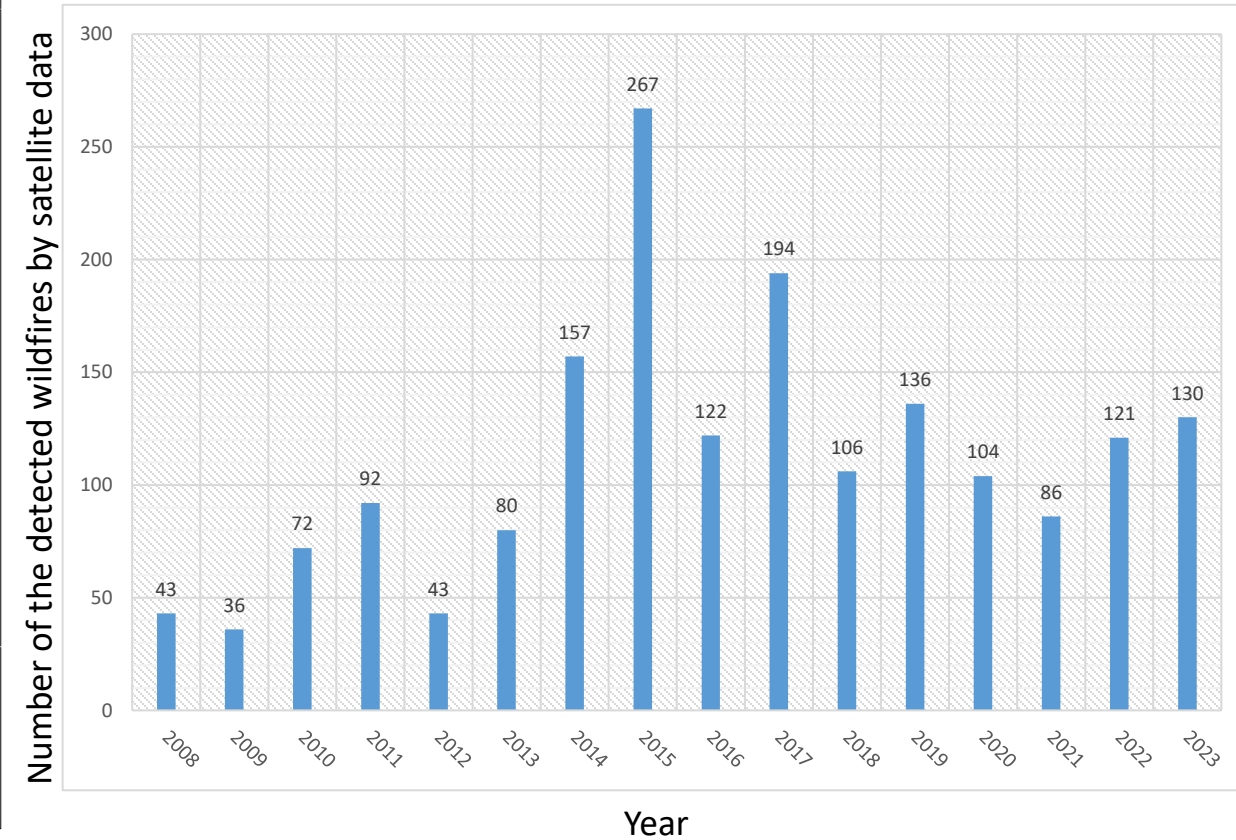
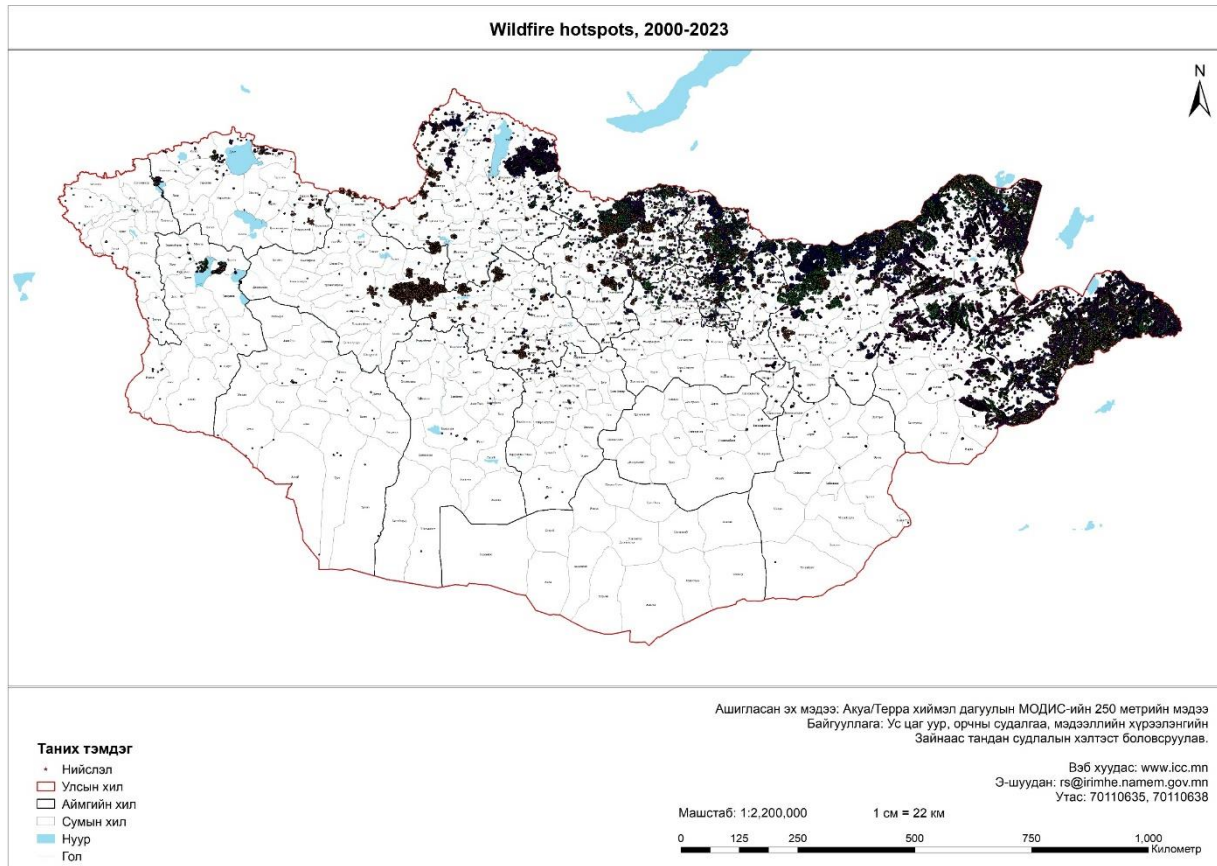
Operational Impact: Higher resolution imaging provides exact coordinates, enabling NEMA (National Emergency Management Agency) to optimize fire suppression deployment. Integrated Terra/Aqua MODIS (1000m/250m) feeds allow for real-time tracking of fire advancement across both forest and steppe terrain.

BOTTOM MAP: BURN SCAR MAPPING - AFFECTED AREA OUTLINE



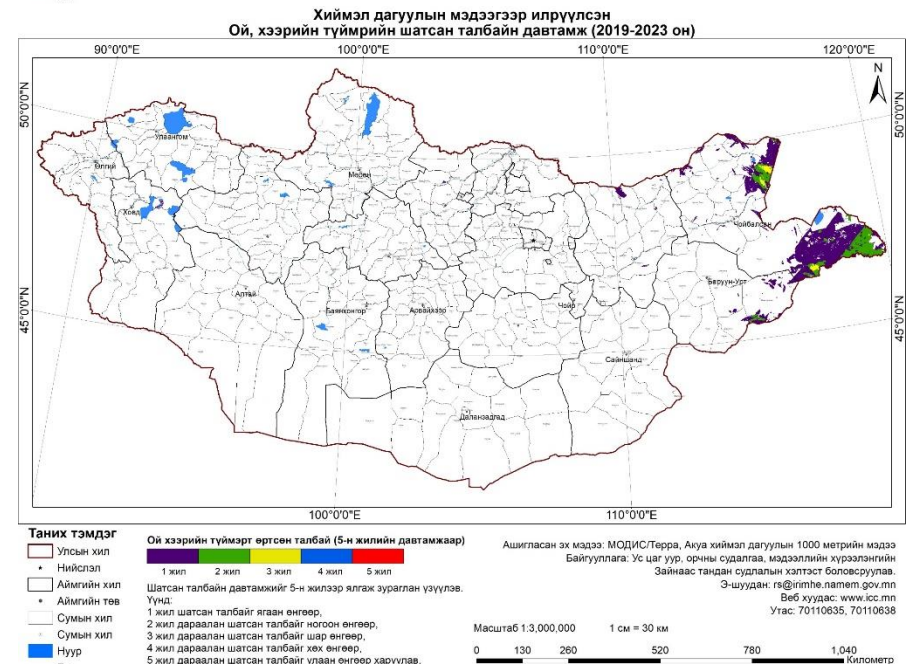
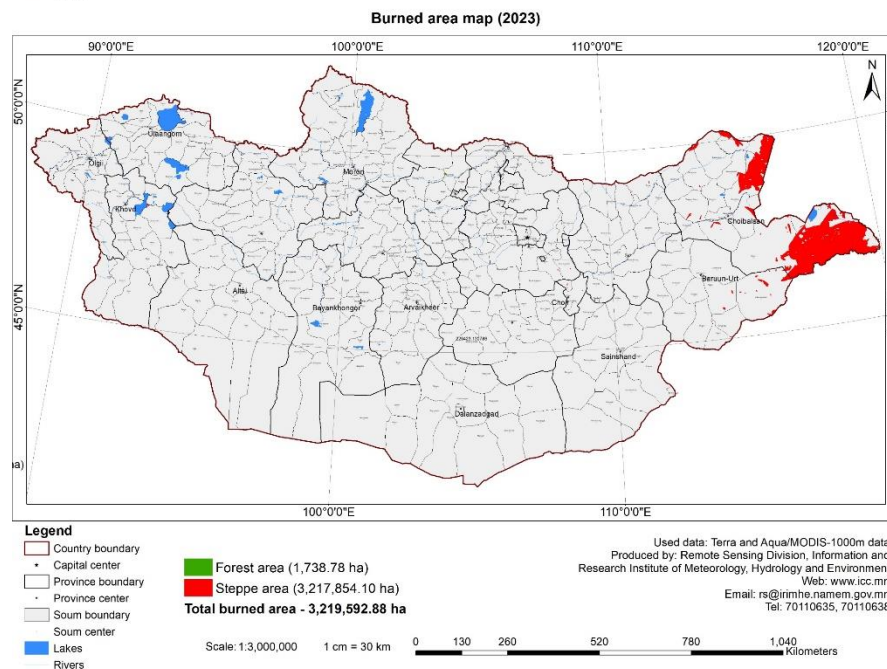
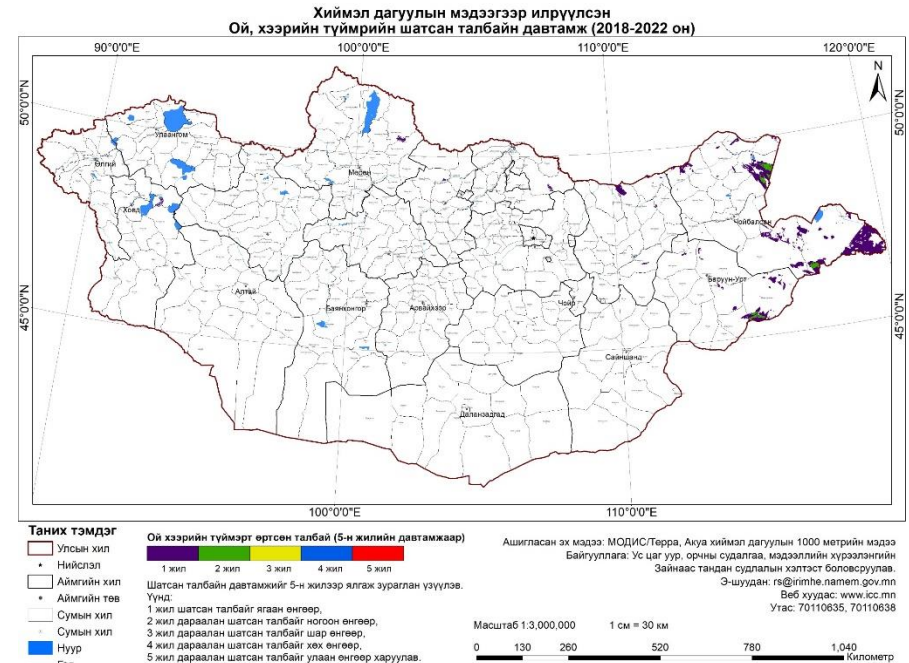
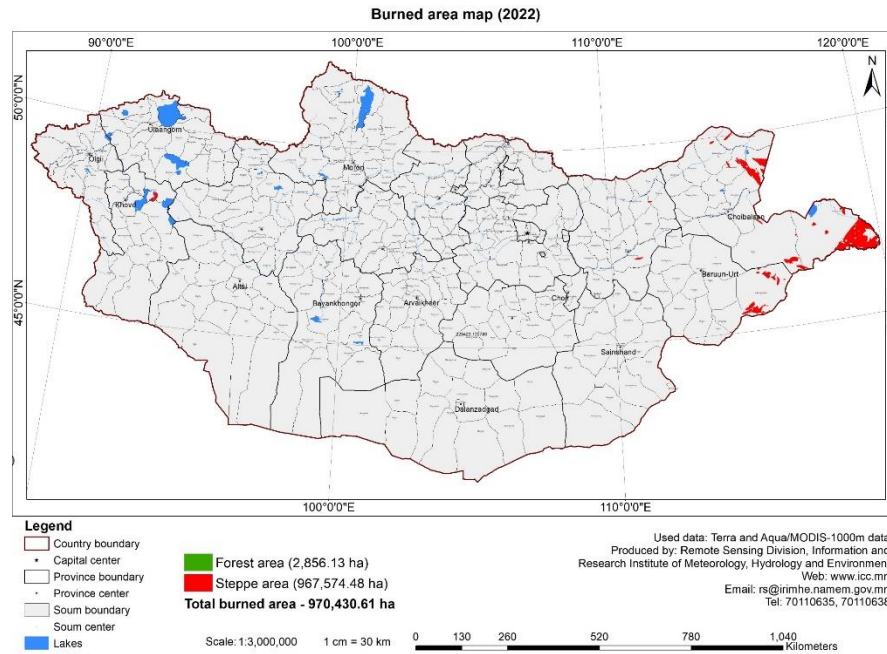
ORBIT: POLAR_SS

REAL-TIME WILDFIRE DETECTION FOR YEARS



Recognition of Disaster occurrence spots Hazard map produced by space-based information

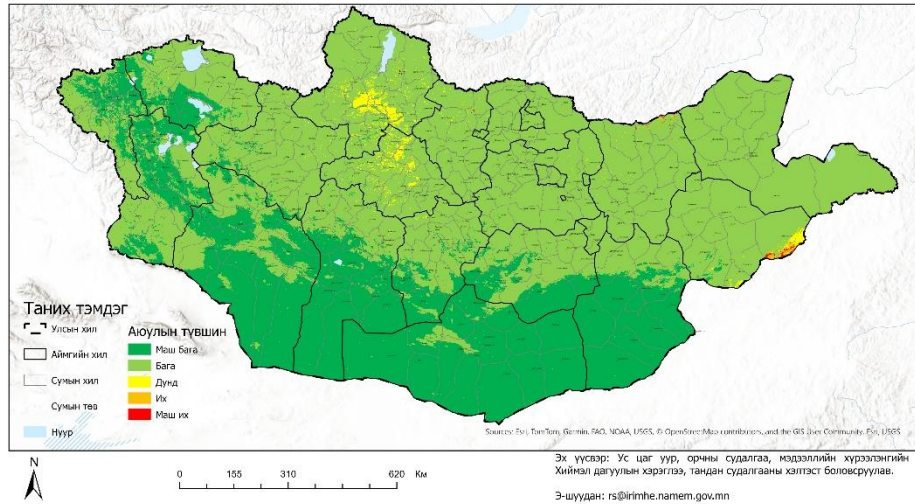
BURNT AREA FOR 2022-2023 AND ITS FREQUENCY FOR 2018-2023



Researches .,

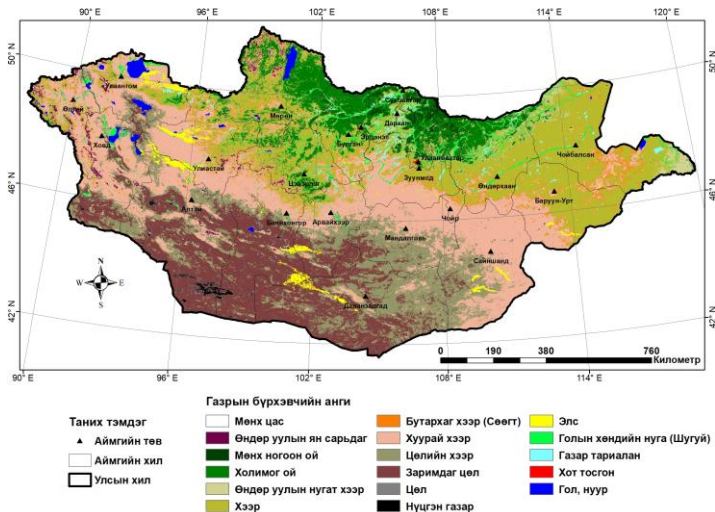
Wildfire hazard mapping

Монгол орны ой, хээрийн түймрийн аюулын зураглал
(2026 оны 3 дугаар сарын 03-ны байдлаар)



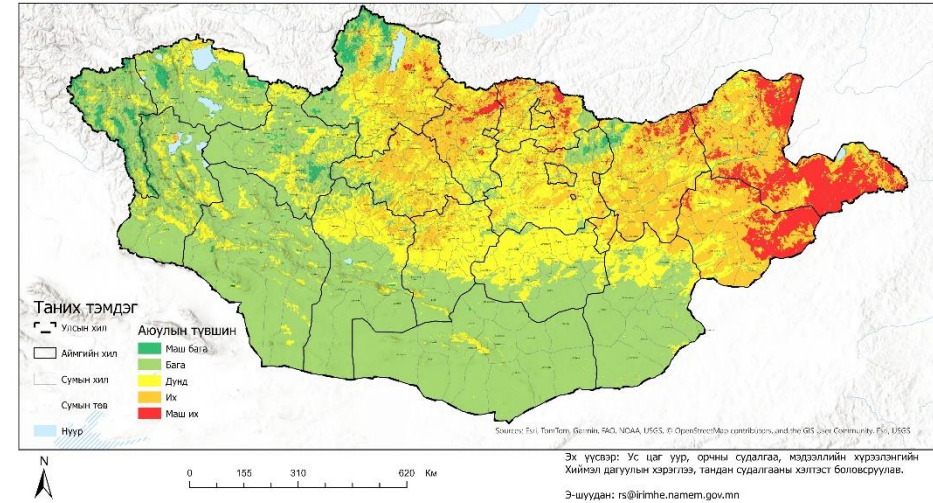
2026.03.03-ны байдлаарх буюу тухайн хугацааны хуурайшилт/цасан бүрхүүлийн нөхцөлд суурилсан Монгол орны ой, хээрийн түймэр гарах байгалийн нөхцөл бүрдсэн эсэхийг харуулсан аюулын зураглал

Монгол орны газрын бүрхэвчийн зураглал



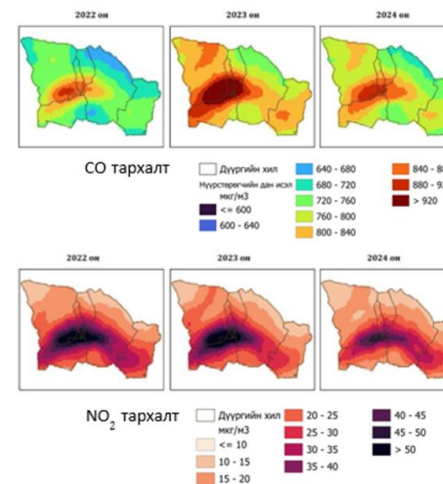
Монгол орны газрын бүрхэвчийг 1995 онд 1992–1993 оны NOAA NDVI өгөгдлөөр газрын бүрхэвчийг 8 ангилалд зурагласан. 2000-2025 онд 5 жил тутам MODIS NDVI өгөгдлөөр 16 ангилалтайгаар зураглаж байна.

Монгол орны ой, хээрийн түймрийн аюулын урьдчилсан зураглал
(2026 оны 4 дүгээр сарын 17–26)



2026 оны 4 дүгээр сарын 17–26-ны хугацааны Загварын үр дүнгээр тооцоолсон хуурайшлын урьдчилсан мэдээнд суурилсан Монгол орны ой, хээрийн түймэр гарах байгалийн нөхцөлийн урьдчилсан төлөв зураглал

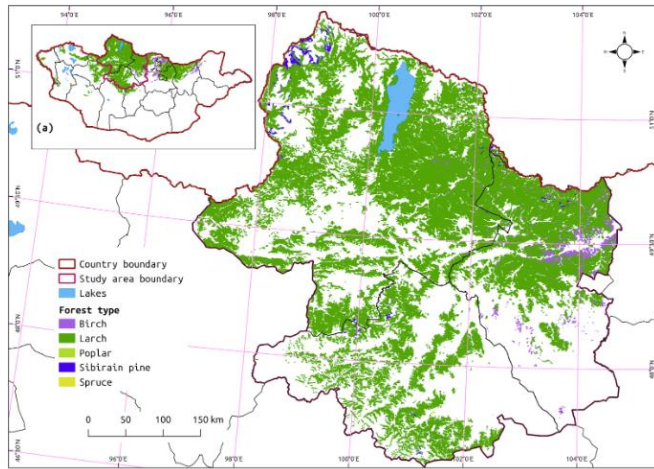
Агаар бохирдуулагч бодисын тархиалтын зураглал



Улаанбаатар хотын агаар бохирдуулагч азотын давхар исэл (NO₂), нүүрстөрөгчийн дутуу исэл (CO) -ийн тархалтыг хиймэл дагуулын мэдээгээр зураглаж байна.

Estimation of aboveground biomass and carbon stock in boreal forest using satellite data with machine learning

Study area



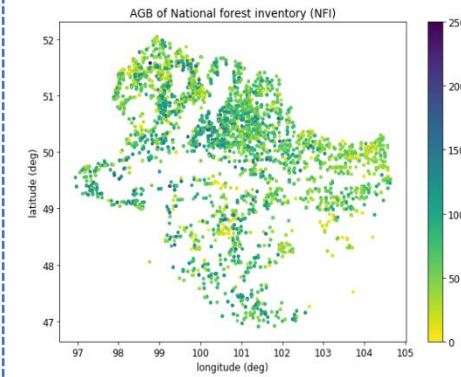
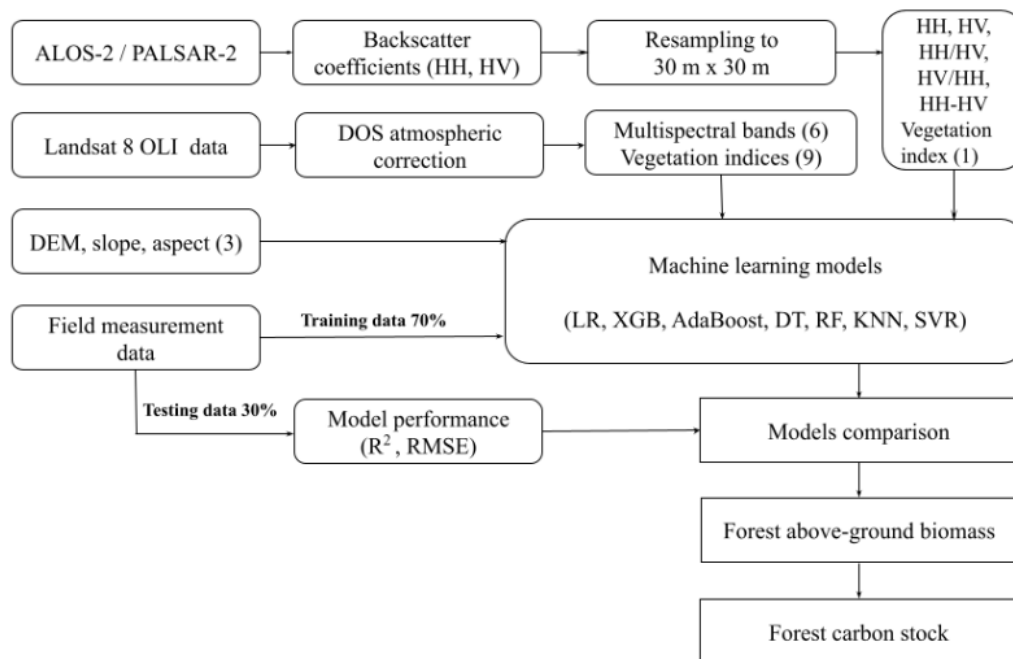
Location: Latitude: 47°N - 53°N, Longitude: 96°E - 104°E

Climate: Average annual rainfall 270 mm, and annual mean temperature around 1.3 °C

Climate classification: Dwc (continental, dry winter, cold summer)

Total area: 20466182 ha

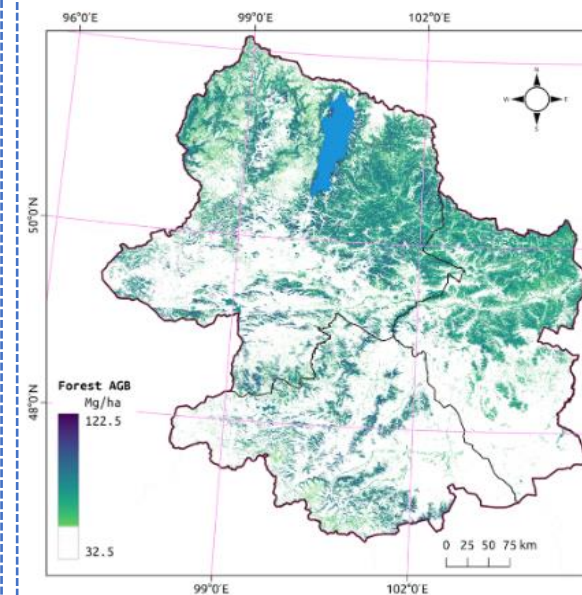
Forested area: 9380532 ha (45.8%)



Number of plots

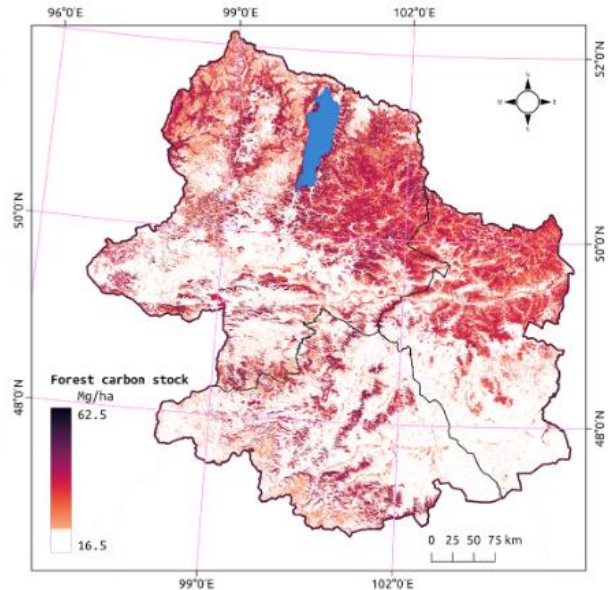
AGB (Mg/ha)	1 - 50	50 - 100	100 - 150	150 - 200	200 - 250
Number of plot	1952	2570	1017	170	11
Average AGB (Mg/ha)	69.7				

Forest above-ground biomass

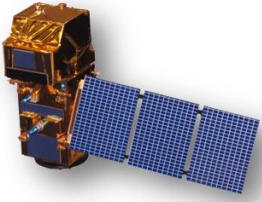


Estimated from Random forest
Average forest AGB: **54.1Mg/ha**

Forest carbon stock



Estimated from AGB
Average carbon stock: **27.5 Mg/ha**



Sentinel 2, 11 years



Terra, Aqua/ MODIS
18 years

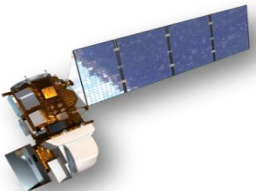
Mongolian Data Cube

A comprehensive Big Data system that consolidates decades of satellite source data, making it ready for analysis.

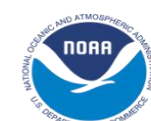


Open access
for all users and
researchers:
datacube.icc.mn

Notebook



Landsat 8, 10 years

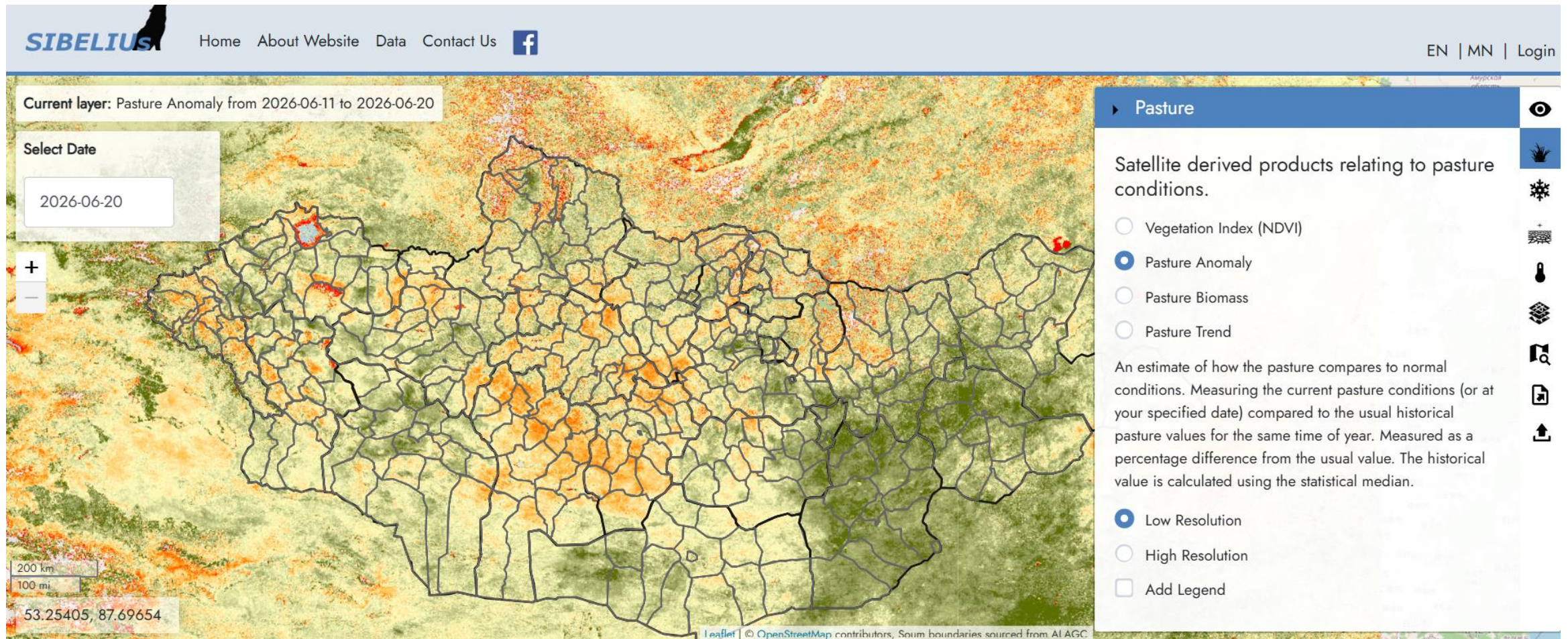


Raytheon



Soumi-NPP VIIRS
(15 years)

The Visualisation Website - [https:// datacube.icc.mn/](https://datacube.icc.mn/)



The Visualisation Website provides a simple way to access useful information and products in the Mongolian Data Cube

Anyone can access the Website

It is available in English and Mongolian



https://en.wikipedia.org/wiki/Project_Jupyter

<https://www.datacube.icc.mn:8000/>

9/14/2026

Access to Mongolian Data Cube

A screenshot of the JupyterHub web interface. The top navigation bar includes the JupyterHub logo, a "Logout" button, and a "Control Panel" button. Below the navigation bar, there are tabs for "Files", "Running", "Clusters", and "Nbextensions". The "Files" tab is active, showing a file browser for the path "/ Users / Elbeg". The file list includes folders like "10day_image", "Anomaly_image", "MODIS_Snow_image", "Pasture Validation", "Sentinel_test", "Training 1", and "Training 2". It also lists various IPYNB files such as "10day_Composite_test.ipynb", "Landsat_true_color.ipynb", "Long-term_NDVI_mean.ipynb", "MODIS_10day_Composite.ipynb", "MODIS_10day_composite_cloudmasked.ipynb", "MODIS_10day_composite_without-cloudmasked.ipynb", "MODIS_daily_cloudmasked_image.ipynb", "MODIS_daily_rgb_image.ipynb", "MODIS_LST-Anomaly.ipynb", "MODIS_LST.ipynb", "MODIS_LST_night.ipynb", "MODIS_NDSI.ipynb", "MODIS_NDVI-Anomaly.ipynb", "MODIS_NDVI.ipynb", "MODIS_pasture.ipynb", and "MODIS_pasture_anomaly.ipynb". Each file entry shows its name, last modified time, and file size. The interface also includes an "Upload" button, a "New" dropdown menu, and a refresh icon at the top right of the file list.

☐ 0	/ Users / Elbeg	Name	Last Modified	File size
		..	seconds ago	
☐		10day_image	7 months ago	
☐		Anomaly_image	a month ago	
☐		MODIS_Snow_image	8 months ago	
☐		Pasture Validation	a month ago	
☐		Sentinel_test	10 months ago	
☐		Training 1	8 months ago	
☐		Training 2	8 months ago	
☐		10day_Composite_test.ipynb	7 months ago	1.62 MB
☐		Landsat_true_color.ipynb	6 months ago	81 kB
☐		Long-term_NDVI_mean.ipynb	6 months ago	14.4 kB
☐		MODIS_10day_Composite.ipynb	6 months ago	1.68 MB
☐		MODIS_10day_composite_cloudmasked.ipynb	24 days ago	821 kB
☐		MODIS_10day_composite_without-cloudmasked.ipynb	7 months ago	531 kB
☐		MODIS_daily_cloudmasked_image.ipynb	7 months ago	493 kB
☐		MODIS_daily_rgb_image.ipynb	17 days ago	99.4 kB
☐		MODIS_LST-Anomaly.ipynb	6 months ago	805 kB
☐		MODIS_LST.ipynb	6 months ago	812 kB
☐		MODIS_LST_night.ipynb	6 months ago	652 kB
☐		MODIS_NDSI.ipynb	6 months ago	45.7 kB
☐		MODIS_NDVI-Anomaly.ipynb	a month ago	342 kB
☐		MODIS_NDVI.ipynb	6 months ago	50.8 kB
☐		MODIS_pasture.ipynb	6 months ago	358 kB
☐		MODIS_pasture_anomaly.ipynb	5 months ago	508 kB



Source: <https://www.youtube.com/watch?v=q4nsm6-K1hA>

lhagvadorj@irimhe.namem.gov.mn